

Problem 7.5

The sum of the present value of 1 paid at the end of n periods and 1 paid at the end of $2n$ periods is 1. Find $(1+i)^{2n}$.

Solution

The present value condition gives

$$\frac{1}{(1+i)^n} + \frac{1}{(1+i)^{2n}} = 1.$$

Combining over a common denominator,

$$\frac{(1+i)^n + 1}{(1+i)^{2n}} = 1.$$

Now let

$$x = (1+i)^n.$$

Then the equation becomes

$$\frac{x+1}{x^2} = 1,$$

so

$$x+1 = x^2.$$

Hence,

$$x^2 - x - 1 = 0.$$

Using the quadratic formula,

$$x = \frac{1 \pm \sqrt{1 - 4(1)(-1)}}{2} = \frac{1 \pm \sqrt{5}}{2}.$$

Since $x = (1+i)^n > 0$, we take the positive root:

$$(1+i)^n = \frac{1 + \sqrt{5}}{2}.$$

Therefore,

$$(1+i)^{2n} = \left(\frac{1 + \sqrt{5}}{2} \right)^2 = \frac{(1 + \sqrt{5})^2}{4} = \frac{6 + 2\sqrt{5}}{4} = \frac{3 + \sqrt{5}}{2}.$$

So,

$$\boxed{(1+i)^{2n} = \frac{3 + \sqrt{5}}{2}}.$$